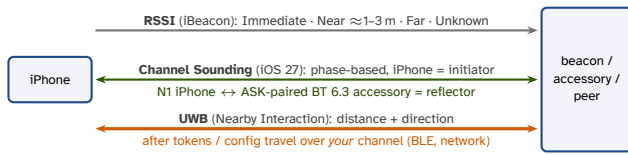


**In one line:** Apple's proximity features are **different frameworks on one radio**: an **iBeacon** is a BLE advert only **Core Location** sees; **AccessorySetupKit** (iOS 18) pairs one accessory without the Bluetooth prompt and now gates Channel Sounding (27) and some background relaunches (26); **Nearby Interaction** measures over **UWB**; **Find My** relays locations encrypted to rotating keys. A plain BLE GATT accessory needs **no MFi**.

An iBeacon advert — 30 of the 31 legacy advertising bytes

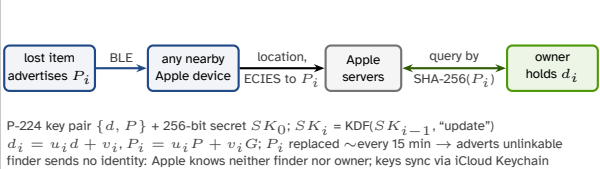
|                    |        |          |              |                                               |                                                                              |  |                    |                          |                         |                                                                                       |
|--------------------|--------|----------|--------------|-----------------------------------------------|------------------------------------------------------------------------------|--|--------------------|--------------------------|-------------------------|---------------------------------------------------------------------------------------|
| 02 01 06           | 1A     | FF       | 4C 00        | 02 15                                         | proximity UUID — 16 B                                                        |  | major 2 B          | minor 2 B                | power 1 B               | no room for a name or service UUID — and iOS never hands this packet to CoreBluetooth |
| Flags AD (type 01) | len 26 | mfr data | Apple 0x004C | 02 = beacon<br>15 = 21 B left<br>(unverified) | the deployer's own value, not registered with Apple; the only required field |  | group (e.g. store) | member (e.g. department) | RSSI at 1 m, signed dBm |                                                                                       |

How far is it? Three radios, three answers

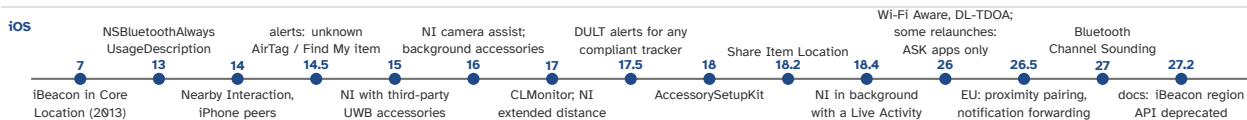


result: `CLLocationDistance` (m), `CLLocationDirection` (unit vector) — nil when not available

Find My: offline finding, end-to-end encrypted



iOS



## Which framework, what it needs

| job                      | framework (iOS)                     | gate                                                                                                                                                                                                                                                                                                                         |
|--------------------------|-------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| GATT, scan, advertise    | CoreBluetooth                       | <code>NSBluetoothAlwaysUsageDescription</code> (13; older targets also <code>_PeripheralUsageDescription</code> ); modes <code>bluetooth-central</code> / <code>-peripheral</code> location authorisation (the same prompt as GPS) app frontmost, screen on, unlocked; <code>withMeasuredPower</code> : nil = device default |
| see a beacon             | Core Location                       | <code>CLBeaconRegion</code> data, <code>CBPeripheralManager</code> <code>AccessorySetupKit</code> (18)                                                                                                                                                                                                                       |
| set up an accessory      | NearbyInteraction (14)              | <code>NSNearbyInteractionUsageDescription</code> ; UWB on both ends (or Channel Sounding, 27); your own token channel                                                                                                                                                                                                        |
| distance, no UWB         | CoreBluetooth Channel Sounding (27) | NI iPhone; ASK-paired, connected BT 6.3 accessory                                                                                                                                                                                                                                                                            |
| lost-item relay MFi link | Find My network                     | MFi Program; Apple's accessory specification                                                                                                                                                                                                                                                                                 |
| P2P Wi-Fi                | Wi-Fi Aware (26)                    | ExternalAccessory <code>UISupportedExternalAccessoryProtocols</code> <code>com.apple.developer.wifi-aware</code> ; <code>WiFiAwareServices</code> ; pair via ASK or <code>DeviceDiscoveryUI</code>                                                                                                                           |

## iBeacon

- Monitoring** (enter / exit, passive, cheap) relaunches a terminated app (location authorisation + the location background mode, per the docs); `CLLocationManager` caps it at **20 regions** per app, shared by all its managers; the same identifier replaces a region; timely events need network connectivity. `CLMonitor` (17): an actor; conditions persist across launches; events is an async sequence; one open instance per name — re-create it after a relaunch. An event = state (satisfied / unsatisfied / unknown) + date; a wildcard match reports the seen major / minor as its *refinement*.
- Ranging**: frequent RSSI → proximity + accuracy (m), for a frontmost app, "not expected in the background". **Immediate** = held up to it; **Far** = seen, confidence too low for Near — not "far away"; **Unknown** = too few samples yet. Beacons arrive sorted by best-guess proximity, and the guess can be wrong.
- Calibrate** every installed beacon: measured power = mean rssi at 1 m over  $\geq 10$  s, sweeping a 30 cm line. Bodies, water and metal attenuate 2.4 GHz; range is tens of metres (geofences  $\sim 100$  m minimum); room-level at best.
- UUIDs can be sniffed by anyone. Building iBeacon hardware takes Apple's licence (the spec comes with it). Docs, Oct 2026: `CLBeaconRegion`, `CLBeaconIdentityConstraint`, `startMonitoring(for:)` deprecated in 27.2 → `CLMonitor` + `CLBeaconIdentityCondition`.

## CoreBluetooth on iOS · the accessory side

- An app advertises only a local name + service UUIDs: 28 B, plus 10 B of scan response for the name; best effort. **In the background**: no name, every UUID in the **overflow area** — seen only by iOS devices scanning for it.
- Relaunch** (TN3115) only while waiting on a Bluetooth event that then happens. Yes: evicted, crashed, restarted (after first unlock), Control Center toggle, Airplane Mode if it switches Bluetooth. No: force quit, Bluetooth off in Settings. **iOS 26**: the force-quit, Control Center and Airplane Mode cases relaunch **only apps that set the accessory up with ASK**.
- Accessory advertising**: 20 ms for  $\geq 30$  s, then *exactly* one of 152.5 · 211.25 · 318.75 · 417.5 · 546.25 · 760 · 852.5 · 1022.5 · 1285 ms (QA1931).
- Connection parameters** (ADG, quoted by Apple, 2025): min  $\geq 15$  ms, a multiple of 15,  $\leq 2$  s; max  $\geq$  min + 15 ms (or both 15); latency  $\leq 30$ ; max  $\times$  (latency+1)  $\leq 6$  s; supervision timeout 6–18 s and  $> 3 \times$  max  $\times$  (latency+1). QA1931 (2017) still says 2–6 s and  $\leq 2$  s.

## AccessorySetupKit (iOS 18)

- `ASAccessorySession.showPicker(for:)` with `ASPickerDisplayItem(name, 180 × 120 pt image, ASDiscoveryDescriptor: service UUID or company id + name substring, data blob + mask, SSID or prefix, range · immediate; Wi-Fi Aware service, 26). One picker at a time.`
- The **system** scans and draws the picker → `ASAccessory` (`bluetoothIdentifier` → `CBPeripheral`, `ssid`). **No Bluetooth or Wi-Fi prompt**; a scan returns only the app's accessories; powered on only once one is paired.
- `Info.plist` `NSAccessorySetupKitSupports` + `NSAccessorySetupBluetoothServices`, `CompanyIdentifiers`, `Names`: discovering anything undeclared **crashes**. Users rename and remove in Settings › Privacy & Security › Accessories. Old pairings: `ASMigrationDisplayItem`, no `CBCentralManager` until done. watchOS 26: the companion watch app may use CoreBluetooth with it, behind its own prompt.

## Nearby Interaction · Channel Sounding

| configuration                                                     | other end                     | bootstrap                                            |
|-------------------------------------------------------------------|-------------------------------|------------------------------------------------------|
| <code>INearbyPeerConfiguration</code> ( <code>peerToken</code> :) | iPhone, Watch                 | <code>NIDiscoveryToken</code> sent over your channel |
| <code>INearbyAccessoryConfiguration</code> ( <code>data</code> :) | UWB accessory (15)            | config data per Apple's NI accessory protocol        |
| ...( <code>accessoryData:bluetooth-PeerIdentifier</code> :)       | BLE-paired UWB accessory (16) | NI GATT service; allows background                   |
| ...( <code>bluetoothChannel-SoundingIdentifier</code> :)          | BT 6.3 accessory (27)         | ASK-paired only                                      |
| <code>NIDLTDAAConfiguration</code>                                | DL-TDOA anchors (26)          | supports <code>DLTDAAMeasurement</code>              |

- `deviceCapabilities` (16, replaces `isSupported`): distance, direction, camera assistance, extended distance (17), DL-TDOA (26), Channel Sounding (27). UWB: iPhone 11 on, not 16e; second-gen chip in iPhone 15 on (not 16e / 17e) and AirTag 2nd gen; barred or limited in some countries. Watch: distance only.
- Camera assistance** (16): ARKit adds `horizontalAngle` + `verticalDirectionEstimate` once the phone has been swept enough.
- A peer that quits or leaves UWB range times out → `session(_:didRemove:reason:)`; a bad token → `invalidConfiguration`; a declined prompt → `userDidNotAllow`.
- Background** suspends the session, unless the peer is BLE-paired and connected (the accessory acts; the app gets no callbacks) or a **Live Activity** starts as the app leaves (18.4); 27.2 docs: DL-TDOA may start on a background relaunch.
- Channel Sounding** (27): `supportsFeatures(.channelSounding)` → `peripheral.startChannelSoundingSession(role .initiator)` → `CBChannelSoundingProcedureResults.distance`; direction through NI + camera. Accessory: Core 6.3 inline PCT, modes 0 + 2,  $T_{FCS} \geq 100 \mu s$ . Foreground only; iOS drops outliers and may sample less under radio load. A Core 6.0 feature; 6.3 shipped May 2026.

## Find My, MFi, EU

- Alerts for an unknown AirTag / Find My item moving with you: 14.5; for any tracker built to **DULT** (Apple + Google, IETF working group): 17.5 and Android 6.0+. **Share Item Location** (18.2): an authenticated link for an airline or anyone, gone after 7 days.
- Of these APIs only `ExternalAccessory` and Find My need **MFi**; a BLE GATT peripheral needs none.
- EU, 26.5**: proximity-triggered pairing (system prompt, then the app finishes in the ASK flow; organisation account, self-certification, companion app) and `AccessoryNotifications` (notifications to a wearable; EU devices + EU accounts only).